(iv) $\exists$ a map $-T_1 : U \to V$ such that

$$\begin{aligned}(T_1 + (-T_1))(u) &= T_1(u) + (-T_1)(u) \\ &= T_1(u) - T_1(u) \\ &= V_1 - V_1 \\ &= 0\end{aligned}$$

(v) $$\begin{aligned}(T_1 + T_2)(u) &= T_1(u) + T_2(u) \\ &= V_1 + V_2 \\ &= V_2 + V_1 = T_2(u) + T_1(u), \\ &= (T_2 + T_1)(u).\end{aligned}$$

(vi) Let $\alpha \in \mathbb{F}$

$$\begin{aligned}\alpha(T_1 + T_2)(u) &= \alpha\{T_1(u) + T_2(u)\} \\ &= \alpha(V_1 + V_2) \\ &= \alpha V_1 + \alpha V_2 \\ &= \alpha T_1(u) + \alpha T_2(u)\end{aligned}$$

(vii) Let $\alpha, \beta \in \mathbb{F}$

$$\begin{aligned}(\alpha+\beta) T_1(u) &= (\alpha+\beta) V_1 \\ &= \alpha V_1 + \beta V_1 \\ &= \alpha T_1(u) + \beta T_1(u)\end{aligned}$$

(viii) $$\begin{aligned}I \cdot T_1(u) &= I(V_1) \\ &= V_1 \\ &= T_1(u)\end{aligned}$$

Since $L(U,V)$ satisfies all the conditions, so $L(U,V)$ is a vectorspace.

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